TITLE: AMAZON ROBOTICS STATION AIR REQUIREMENTS		
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DEPARTMENT: HW ENGINEERING		PAGE 1 OF 8

CONFIDENTIAL AND PROPRIETARY INFORMATION

AMAZON ROBOTICS LLC

300 RIVERPARK DRIVE

NORTH READING, MA 01864

REVISION HISTORY

Rev.	Description of Change	Sections Affected	Originator/Department
01	Initial release of station air requirements	ALL	T. Lilliston / Engineering
02	Reduced air flow from 5 to 1 cfm and changed purity class from 2 to 6	5.1 & 6.1	T. Lilliston / Engineering
03	Added LOTO and WWDE to acronyms list Rewrote entire section to include details of air distribution system requirements, air drop height, and LOTO requirements Added Cardinal air requirements Added Sparrow air requirements Added new Figure 2 showing LOTO device on ball valve	4.0 5.0 7.0 8.0 9.0	M. DellaCroce / Engineering 08/01/2022
04	Added “The compressed air drop location to Cardinal Added TCC air requirements Added Vulcan air requirements	7.1 9.0 10.0	M. DellaCroce / Engineering 04/20/2022
05	Added vacuum requirements for Cardinal Updated Sparrow compressed air demand requirements Updated TCC compressed air demand requirements Removed Vulcan from the specification.	7.2 8.1 9.1 10.1	K. Bucher/Engineering 02/21/25

Table of Contents

Table of Figures

1 Overview

The purpose of this document is to provide station air requirements for an Amazon Robotics Mobile-robotic Fulfilment System installation and Robotic Sort Centers. Where just the word “air” is used, it can be assumed to apply for both vacuum and compressed air applications. Amazon Robotics will not provide or install any of the materials indicated by this document unless otherwise noted.

The requirements pertain to station air requirements for the following:

- Amazon Robotics Stations and Work Cells

2 Reference Documents

The reference documents listed below are listed for traceability purposes to future modifications of this station air requirements as they contain relevant information to the requirements listed herein. Readers of this document do not need access to these references to complete a compressed air installation.

- ISO-8573.1

3 Relevant Codes

The requirements detailed in this document are intended to be in compliance with all local and national building codes as applying to each individual building location. If local or national codes are more stringent that requirements specified herein, the supplier / contractor shall ensure compliance with the local / national legal requirement. In the absence of a relevant local/national code, the contractor shall determine the appropriate standards or norms to be followed, subject always to the client’s approval. The contractor should also satisfy themselves as to any additional requirements as may be laid down by the client in their tender package.

In the event of a conflict between the requirements contained within this document and those laid out under local or national code, (or indeed those as set out elsewhere by the client), then Amazon Robotics must be notified in writing prior to any design or installation work being undertaken for any items contained herein.

4 Acronyms

- **AFT** – Amazon Fulfillment Technology
- **BOM** – Bill of Materials
- **DU** – Drive Unit
- **LOTO** – Lock out tagout
- **TCC** – Transfer Cartesian Cell
- **WWDE** – World Wide Design and Engineering

5 General Requirements

Unless otherwise noted these requirements apply to all compressed air drops.

General Requirements/Specifications	
5.1 ***Air Drops – Installation and System Requirements***	1. Air piping shall be adequately supported to appropriate building members to eliminate strain on system piping.
	2. Compressed air shall be provided as clean dry air in accordance with ISO-8573-1 purity Class 6.
	3. Air drops shall be provided with a local approved shutoff ball valve to facilitate maintenance at station. a. The local approved shutoff ball valve shall be compatible with a LOTO device for maintenance. See Figure 2 for image of LOTO device.
	4. The compressed air distribution system supplying Amazon Robotics fulfillment and sortation systems air drops shall be in accordance with the applicable requirements of WWDE compressed air distribution system requirements found in the Material Handling Applications Standards of the Amazon Project and Equipment Standards.
5.2 ***Air Drops - Labeling***	1. Labels must be sufficiently durable to withstand a 15 second mineral spirit rub, followed by a 15 second water rub with a soft cloth test. Labels must remain legible with no curling of edges permitted.
5.3 ***Air Drops - Location***	a. For <u>Amazon Robotics Stations or Work Cells</u>, the location of the **air drops** shall be as indicated on the applicable interface control drawings and station assembly drawings. b. **NOTE:** The height from the finished floor to the bottom of the shutoff ball valve shall be between 36-inches to 67-inches for ease of access to the shutoff ball valve, unless otherwise stated

6 Robin

This section provides compressed air requirements for Robin.

Robin Station Compressed Air Specifications	
6.1 ***Individual Drop***	1. One compressed air drop shall be supplied per Robin station capable of supplying 100 psig of clean dry air at a rate of 1 CFM. Maximum pressure not to exceed 140 psig. The interface shall include an 8mm push to connect air fitting with an approved shutoff valve upstream of the fitting to facilitate station maintenance.

7 Cardinal

This section provides compressed air requirements for Cardinal.

Cardinal Station Compressed Air Specifications	
7.1 ***Individual Drop***	1. One compressed air drop shall be supplied per Cardinal station capable of supplying 100 psig of clean dry air at a rate of 1 CFM. Maximum pressure not to exceed 140 psig. The interface shall include a 12mm push to connect air fitting with an approved shutoff valve upstream of the fitting to facilitate station maintenance. 2. The compressed air drop location shall be 110-inches above the finished floor and located no more than 6-inches in plan view from the survey point location. This is an isolation valve, LOTO requirements are to follow Chapter 5.

Cardinal Station Vacuum Air Specifications	
7.2 ***Individual Drop***	1. One vacuum air drop shall be supplied per Cardinal station capable of supplying vacuum at -12psig at a flow rate of 4000 SCFH. Average station usage shall not exceed 3400 SCFH. Vacuum pressure shall not be less than -13.5psig. The interface shall include a 1-1/2” hose barb fitting with an approved isolation valve upstream of the fitting to facilitate station maintenance. 2. The vacuum air drop location shall be 110-inches above the finished floor and located no more than 6-inches in plan view from the survey point location. This is an isolation valve, LOTO requirements are to follow Chapter 5.

8 Sparrow

This section provides compressed air requirements for Sparrow.

Sparrow Station Compressed Air Specifications	
8.1 ***Individual Drop***	1. One compressed air drop shall be supplied per Sparrow station capable of supplying 100 psig of clean dry air at a rate of 4.3 CFM. Maximum pressure not to exceed 140 psig. The interface shall include a 16mm push to connect air fitting with an approved shutoff valve upstream of the fitting to facilitate station maintenance. 2. The compressed air drop location shall be 110-inches above the finished floor and located no more than 6-inches in plan view from the survey point location. This is an isolation valve, LOTO requirements are to follow Chapter 5.

9 TCC

This section provides compressed air requirements for TCC.

TCC Compressed Air Specifications	
9.1 ***Individual Drop***	1. One compressed air drop shall be supplied per TCC station capable of supplying 101.5 psig of clean dry air at a rate of 2.8 CFM CFM. Maximum pressure not to exceed 140 psig. The interface shall include a 10mm push to connect air fitting with an approved shutoff valve upstream of the fitting to facilitate station maintenance. 2. The compressed air drop location shall be 110-inches above the finished floor and located no more than 6-inches in plan view from the survey point location. This is an isolation valve, LOTO requirements are to follow Chapter 5.

FOR REFERENCE ONLY

10 Example Compressed Air Drop Installations

Figure 1- Air Drop with local approved shutoff ball valve wall mounted with support clamps LOTO device not shown for image clarity

Figure 2- Air drop with local approved shutoff ball valve with LOTO device